

Paralytic Squint

TYPES OF CASES:

- Oculomotor Nerve Palsy
- Trochlear Nerve Palsy
- Abducent Nerve Palsy

HISTORY:

Presenting Complaints:

- Diplopia
- Pain
- Drooping of Eyelid

Details:

Diplopia (ptosis may conceal diplopia)

- Onset
- Unilateral or Bilateral
- Vertical or Horizontal
- Worsening or Stationary
- Diurnal Variation
- Maximum in which field of gaze

Pain

- Onset and relationship with diplopia
- Severity
- Localisation

Drooping of eye lid

- Onset
- Progression
- Diurnal Variation
- Association with fatigue

Visual Complaints

- s/o Cavernous Sinus Thrombosis: Redness, Proptosis

CNS History

- Headache, Nausea, Vomiting (also: Neck rigidity)
- Loss of consciousness, Convulsions
- Focal Neurological Defects or TIA (Hemiparesis, Hemi-anaesthesia)
- Aphasia, Visual Hallucinations
- Ataxia, Dysarthria, Dysphagia

Past History

- Diabetes Mellitus, Hypertension
- Ischaemic Heart Disease, Rheumatic Heart Disease
- Trauma (head injury or orbital trauma)
- Hearing Loss, Tinnitus, Ear Infection, Sinusitis, Epistaxis
- Malignancies: Weight loss, Neck surgeries, Radiation therapy
- Hyperthyroidism: Tremors, Palpitations, Insomnia
- Tuberculosis
- Child birth (Pituitary Apoplexy)

Personal History

- Alcohol, Tobacco smoking
- IV Drug abuse

Causes of Ophthalmoplegia

- Congenital
- Traumatic
- Neoplastic
- Demyelinating Disease
- Inflammatory
- Ischaemic
- Toxic
- Idiopathic

Painful Ophthalmoplegia:

- Intracranial Aneurysm
- Cavernous Sinus Thrombosis
- Orbital Apex Syndrome
- Tolosa Hunt Syndrome
- Orbital Cellulitis
- Pseudotumor of the Orbit

Mimics of cranial nerve palsies:

3rd Nerve and 4th Nerve:

- Thyroid Eye Disease
- Myasthenia Gravis

6th Nerve:

- Thyroid Eye Disease
- Myasthenia Gravis
- Medial Wall Blow-Out
- Orbital Myositis
- Convergence Spasm
- Eso-Duane's Syndrome
- Infantile Esotropia

Examination:

- General Physical Examination
- CNS Examination:
HMF, Cranial Nerves, Sensory, Motor – gait, power, reflexes, coordination

OCULAR EXAMINATION:

Head posture:

- Head tilt
- Chin elevation
- Face turn

Ocular Posture:

- General Ocular Inspection (Exo- or enophthalmos)
- Hirschberg's Test
- Cover/Uncover test
 - Tropia or Phoria
 - Concomitant or Incomitant (Primary and Secondary deviation)
 - Nystagmus: type, jerky or pendular, direction, null point
- Diplopia charting (Hess chart or Lees screen if possible)
- Park-Bielschowsky's 3-step test for vertical deviation
- Detect cyclo-deviation (Double Maddox)

Ocular Examination

- Visual Acuity (Best corrected for distance and near, Colour, Confrontation)
- Lids
 - Palpebral fissures (equal, ptosis)
- Anterior Segment Examination (aniridia, coloboma, heterochromia, cataract)
- Pupil: direct and consensual (comment on near reflex if reaction to light is poor)
- Fundus (Distant Direct Ophthalmoscopy, Direct and Indirect Ophthalmoscopy)
 - Disc
 - Size, Shape, Margins, C/D ratio (Horizontal and Vertical)
 - Neuro-retinal rim (colour and thinning, capillary count)
 - Spontaneous Venous Pulsations
 - Peripapillary defects, NFL defects
 - Retinal vessels, fundus periphery and macula
 - Remember:
 - Bruckner's Test
 - Fundus Torsion
 - Eccentric Fixation (Foveal, Parafoveal, Perifoveal)
- 4. Extraocular Movement
 - Duction, Versions, Convergence (watch for inhibitional palsy)
 - Saccadic Velocity
- 5. Refraction

Intraocular Pressure (Differential Tonometry for muscle restrictions)

Past Pointing:

This test indicates a recent onset. Elicited by asking the patient to point to an object with his finger while viewing it through the paretic eye. The patient will initially point further than the object.

* Important to look out for internuclear ophthalmoplegia and other gaze palsies

Diplopia Charting:

1. Maximum separation is in the direction in which the muscle acts the most
2. The image that appears farthest belongs to the deviating eye
3. The image is displaced in the direction of the paralysed muscle

Watch for Pseudoptosis!

Associated with hypotropia and can be mistaken for ptosis of 3rd CN Palsy.

Internal Ophthalmoplegia:

Isolated involvement of accommodation and pupillary reflex – occurs with botulism and diphtheria.

Pupil Sparing 3rd Nerve Palsy:

Ischaemic mononeuropathy due to diabetes mellitus, hypertension, etc. (**Medical** 3rd Nerve Palsy)

Miosed Pupil in 3rd CN Palsy:

- Associated Horner's Syndrome
- Hutchinson's Pupil in Evolution

Stages of a Paralytic Squint:

1. **Paresis** of Muscle
2. **Overacting** ipsilateral antagonist
3. **Inhibitional Palsy:** Underacting antagonist of contralateral synergist
4. Long standing cases: Comitance (aka: **spread of comitance**)